

Joanne Hoar

Teaching & Education

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Summary

Technology Educator with 20+ years of software development expertise and 2+ years of formal teaching experience. Proven track record delivering engaging, real-world programming instruction across full-stack development, data analytics, and advanced topics. Strong mentoring background with positive student feedback and curriculum development skills.

Education

Master's Degree, Computer Science , University of Calgary <i>Thesis: Agent-based Development of Natural Transportation Networks</i>	2002 – 2004
Bachelor's Degree, Computer Science , University of Calgary	1998 – 2001

Experience

Technology Instructor , <i>Robertson College (contract)</i> <i>C#, .NET, ASP .NET Core, Java, JavaScript, React, Front End Frameworks, Git, GitHub, Azure, SQL, Testing and QA, Data Collection</i>	Jan 2025 – present
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- Instructed 20 course offerings across 12 subjects for Full Stack Development (FSD) and Data Analytics (DA) diploma programs.
- Courses taught: C# Programming, .NET, ASP .NET, ASP .NET Core, Object-oriented Programming (Java), Front End Frameworks, Intro to JavaScript, Web Development Basics, Data Collection, Testing and Quality Assurance, Version Control with Git and GitHub, Introduction to IT.
- Deliver a minimum of 3 synchronous live sessions per week per course, hosting live coding sessions and debugging student projects in real-time.
- Manage and maintain a GitHub repository hosting open-source course materials and sample code.
- Archive session recordings for asynchronous review and participate in ongoing monthly instructor training seminars.
- Maintained an average student satisfaction rating of 4.5/5.0 across evaluated courses, with feedback highlighting approachability, real-world teaching approach, and responsiveness.

Co-founder / Lead Developer / Instructor , <i>Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.)</i> <i>Unreal Engine, Objective-C, Swift, iOS, Android Native, Kotlin, Augmented Reality, ARKit, C++, OpenGL, PHP, SQL, MacOS, Windows, Linux</i>	2005 – 2022
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- Designed, coded, and published eleven games to commercial app stores for mobile devices “including Star Chase and Bitsy (2008, physics-based iOS games released under the SuRJE name) and nine further titles (2011–2022) under the Silicon Hanna brand.
- Developed course curriculum and instructed intermediate C++ programming for the University of Calgary Department of Continuing Education; prepared well-organised lectures, sample code, and assignments; received positive student feedback.
- Technical writing and documentation (including for the Calgary Board of Education), web development with Google Maps integration and SEO for real-estate clients, Firebase, and server administration.
- Designed, hosted, and maintained the website for the Business and Professional Women's Club of Calgary (2009).
- Explored cutting-edge device technologies and SDKs including ARKit augmented reality, Chipmunk physics, and Unreal Engine.
- Developed two unreleased titles: Climate Crisis, an old-west resource-gathering economy-building open-world game starring historical Albertan cowboy John Ware; and Cubey, an open-world math-learning game with an original story in which the character Zero saves the world.

Web Developer / VR Artist (Contract) , <i>Nickle Arts Museum, University of Calgary “collaboration with artist Gerald Hushlak</i> <i>C++, OpenGL, CAVE VR, 3D Scanning, JavaScript, PHP, HTML, CSS</i>	2002 – 2004
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- Gathered requirements, designed, and implemented the interactive online catalogue for the Gerald Hushlak art exhibit (swarmart.com) â€” built interactive features using JavaScript.
- Contributed original generative art to the exhibit: Bacterial Chemotaxis visualisations produced from simulation research were displayed alongside the featured works.
- Collaborated with Hushlak on a pioneering VR installation using the University of Calgary CAVE (Cave Automatic Virtual Environment) â€” 3D-scanned the artist's head, loaded the resulting models into the CAVE, and built an immersive spatial experience years before consumer VR existed.

Research Assistant, *VLab Computer Graphics Group / Artificial Intelligence Research Hub* 2001 – 2004
C++, OpenGL, Solaris, MacOS, Unix, Linux, PHP, MySQL, Qt

- Developed graphics software tools used fractal math and biological modelling.
- Managed network setup and code base for distributed computing.
- Participated in interdepartmental projects and published five research papers.
- Served as teaching assistant and lab instructor across three courses: Introduction to Programming (first year), Computing for Non-Majors (first year), and Operating Systems (third year) â€” led weekly lab sessions, supplemented instructor lesson plans with independent presentations and code and mathematics walk-throughs, and received positive student feedback.

Honours & Awards

Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies	2005
Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada	2005
Graduate Student Scholarship, Alberta Learning	2004
Best Student Paper Award (Co-author), ACAL	2003
CPSC Departmental Research Award, University of Calgary	2003
Graduate Student Conference Travel Grant, University of Calgary Research Services	2003
Academic Travel and Conference Funding, University of Calgary Graduate Students' Association	2003
Academic Travel and Conference Funding, University of Calgary Students' Union	2001

Portfolio

 [xcodegirl/career](https://github.com/xcodegirl/career)